

Final Product Requirements Document (PRD)

Investment Management System for Small Family Office

Version: 1.0 – January 2026

Status: Approved for MVP Implementation (Scenario 1 – Minimal Extension)

1. Introduction

1.1 Overview

This is a hybrid AI-powered investment management platform designed for a small family office with 4 medium-tech-savvy users. The system centralizes information across approximately 8 legal entities (2 Irrevocable Trusts, 3 LLCs, 2 Personal Accounts) and 50–100 investments. It automates document and communication ingestion, performs multilingual extraction while preserving original language, proposes summaries and actions, requires explicit user approval for all critical changes, and supports broad natural-language queries.

Primary interfaces:

- **Slack** – Real-time ingestion, notifications, HDL reviews, quick queries, and approvals.
- **Web** – Visual dashboards, detailed views, document access, and deeper analysis.

The system builds directly on the existing **gabriel-agent** codebase and Google Cloud infrastructure (Cloud Run, Cloud SQL, Cloud Storage, Secret Manager, Drive/Gmail APIs) with minimal extensions to avoid months of development.

1.2 Goals & Success Criteria

- Reduce administrative time on document/communication management by $\geq 50\%$.
- Eliminate missed obligations and deadlines.
- Achieve 100% user adoption within 1 month of launch.
- Maintain data accuracy through human-in-the-loop (HDL) approvals.

1.3 Scope

In Scope

- Fresh-start implementation (no import of old data).
- Hybrid Slack + web UX.

- Multilingual handling with original-language fidelity.
- All critical actions (status changes, task creation, resolutions) require user confirmation.

Out of Scope

- Automatic triggers for status changes or task creation.
- Advanced financial modeling or live market feeds.

2. User Personas

1. **Investment Manager** – Primary oversight; uses web dashboards and Slack for quick checks.
2. **Administrative Assistant** – Handles organization and reminders; heavy Slack user.
3. **Family Principal** – High-level reviews; prefers web visuals.
4. **Advisor** – Occasional view-only access (future).

All users have full visibility across entities and investments.

3. Core Data Model (Cloud SQL – PostgreSQL)

Entity	Key Fields
Entity	id, name, type (Trust/LLC/Personal), ownership_split, formation_docs_link, ein_tax_id
Investment	id, short_name, fund_operator, entity_id (FK), investment_id, commitment_amount, subscription_date, amount_called, category, status (Potential/Active/Closed), expected_irr, actual_irr, interest_rate, notes
Document	id, type, google_drive_link, upload_date, extracted_fields (JSON), two_line_summary (original language), tags, investment_id (FK)
Task/Obligation	id, description, due_date, status, frequency (recurring), reminder_settings, investment_id (FK), linked_document_id (FK)

RAG search uses FAISS vectors persisted in GCS bucket gabriel-agent-faiss.

4. Functional Requirements & User Stories

4.1 Investment & Entity Management

1. As any user, I want investments clearly organized under their owning entity with ownership details visible.

2. As any user, I want proposed investment status changes presented as approvable actions (with supporting AI summary, extracted data, and document link) requiring my explicit confirmation.

4.2 Document Ingestion & Processing

3. As any user, I want new documents/emails automatically ingested from Gmail inbox or Google Drive bucket folders with fuzzy matching and $\geq 90\%$ confidence auto-assignment to entity/investment.
4. As any user, I want OCR and multilingual extraction that preserves original text (e.g., Spanish mutuo clauses, English EV3 tables, Strobe metrics).
5. As any user, I want two-line summaries generated in the document's original language and displayed in Slack channels and web views.

4.3 AI Interaction & Human-in-the-Loop (HDL)

6. As any user, I want to ask broad natural-language questions (e.g., "Compare actual vs expected performance in crypto category") and receive responses simultaneously in a web dashboard view and Slack thread.
7. As any user, I want low-confidence assignments, inconsistencies, or proposed actions presented via HDL with original-language snippets, document links, and AI-suggested resolutions, awaiting my confirmation before proceeding.

4.4 Tasks, Obligations & Reminders

8. As an Administrative Assistant, I want recurring obligations (e.g., quarterly interest from mutuo) auto-suggested based on extracted patterns and proposed for my confirmation.
9. As any user, I want searchable, threaded communication and document histories synced between Slack and web.

4.5 Dashboards & Reporting

10. As any user, I want web-based dashboards with tables and pie charts showing portfolio performance (expected vs. actual IRR, commitment vs. called amounts) by entity, category, and shareholder.
11. As any user, I want one-click access to the full original document via Google Drive link from dashboards, Slack, or AI responses.

4.6 Migration & Fresh Onboarding

12. As an admin user, I want a standalone fresh-start migration tool that imports a master CSV to create entities/investments, batch-processes existing documents with fuzzy matching, resolves issues via HDL, and produces an importable report.

5. Non-Functional Requirements

- **Performance:** Support 100 investments + 1,000 documents with <2s response times.
- **Security:** All secrets via Secret Manager; audit logs for approvals; IAM roles as currently configured.
- **Reliability:** All critical changes require explicit user approval; fallback to HDL on <90% confidence.
- **Usability:** Intuitive hybrid experience; original-language fidelity in summaries/reviews.

6. Technical Foundation (Scenario 1 – Minimal Extension)

- **Backend:** Existing Python/LangChain/OpenAI + Google Cloud Vision API enabled for multilingual OCR.
- **Database:** Existing Cloud SQL PostgreSQL instance (extend schema via migrations).
- **Vector Search:** Existing FAISS with GCS persistence.
- **Deployment:** Current Cloud Run service (update Docker/image via existing cloudbuild.yaml).
- **Web UI:** Add lightweight Streamlit dashboard hosted on the same Cloud Run instance.
- **Integrations:** Existing Slack service + Drive/Gmail APIs; no new major services.

This approach reuses ~95% of current infrastructure and code, enabling MVP delivery in weeks rather than months.

7. Approval & Next Steps

Approved for Implementation – Scenario 1 (Minimal Extension)

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Next: Roadmap & MVP Implementation Plan (Slack-first features → Web dashboards → Migration tool).

This document is now the single source of truth moving forward. Save and version-control it in your repo. Ready when you are to proceed to the roadmap!